

UNIT 1
MODULE 1: FUNDAMENTALS IN CHEMISTRY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED PRACTICAL ACTIVITIES
4. <u>Redox Reactions</u> Students should be able to: 4.1. explain redox reactions in terms of electron transfer and changes in oxidation state (number); 4.2. construct relevant half equations for redox reactions; 4.3. deduce balanced equations for redox reactions from relevant half equations; and, 4.4. order elements in terms of oxidising or reducing ability.	Refer to Module 1, Specific Objective 3.8. <i>Redox equations should be constructed under both acidic and basic conditions.</i>	Perform simple displacement reactions to order elements in terms of oxidising or reducing ability; addition of zinc to copper (II) sulfate solution; addition of chlorine water to bromide or iodide solutions.
5. <u>Kinetic Theory</u> Students should be able to: 5.1. state the basic assumptions of the kinetic theory with reference to an ideal gas; 5.2. explain the differences between real and ideal gases;	Qualitative treatment only – the conditions which are necessary for a gas to approach ideal behaviour, the limitations of ideality at very high pressures and very low temperatures. Include graphical representations.	

